

Grow And Feeds Patrons



Automated Organic Waste Logistics

Bridging the gap between waste generators and sustainable Black Soldier Fly processing using WhatsApp and AI.



The Problem

Manual waste collection is fragmented, causing environmental degradation and supply chain inefficiency.

Smallholder farmers and market vendors lack a low-friction organic waste disposal channel.

BSF processing facilities face inconsistent feedstock supply from manual coordination.

Logistical friction causes waste spoilage, methane emissions, and lost rural economic value.



The Solution

A zero-friction, **WhatsApp-based CRM** that automates intake, dispatch, and **patron communication**.

Conversational Intake:

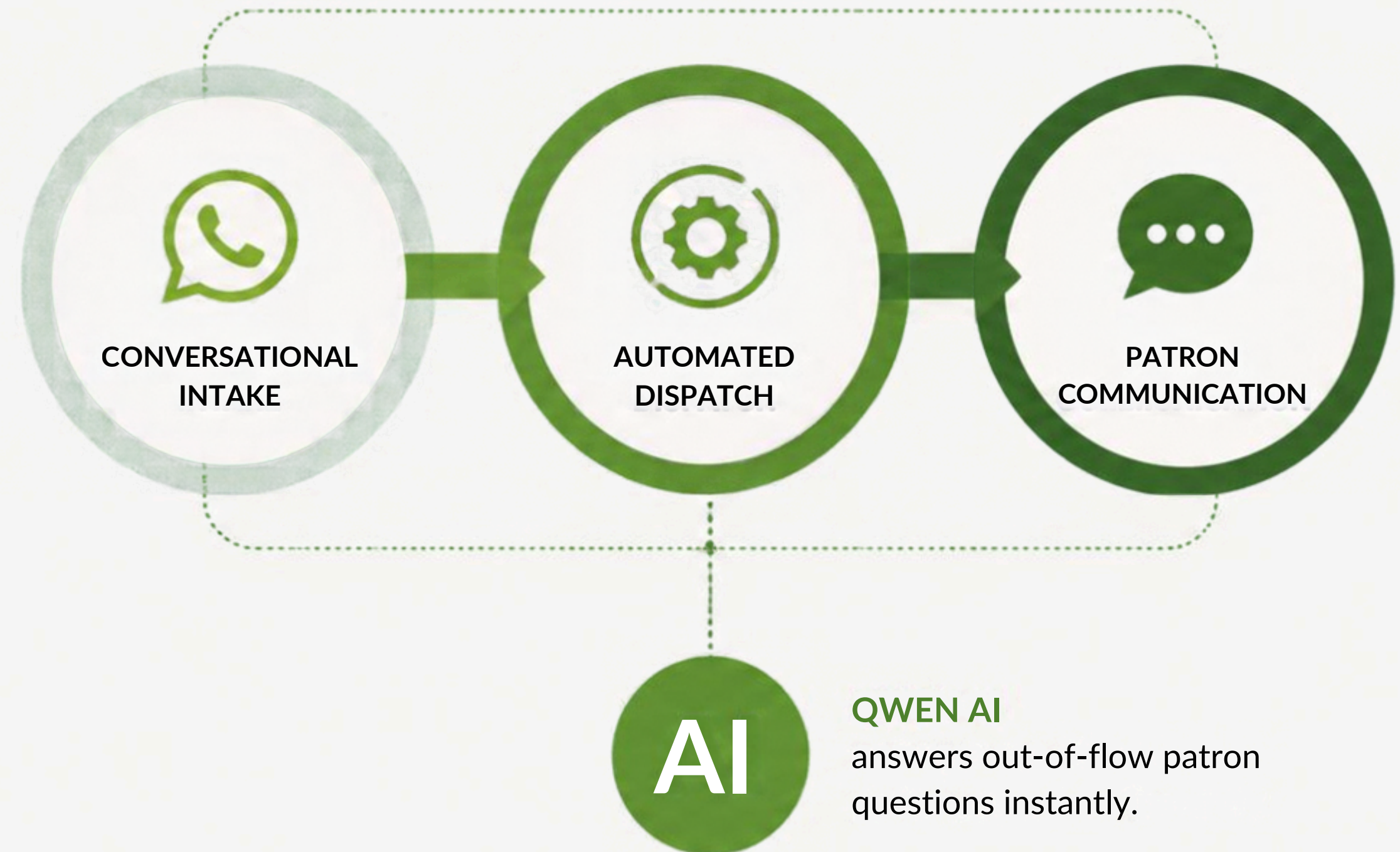
6-step WhatsApp flow capturing waste details without app installs.

Three-Step Transparency:

automated updates at assignment, collection, and processing.

AI-Powered Engagement:

Qwen AI answers out-of-flow patron questions instantly.



System Architecture

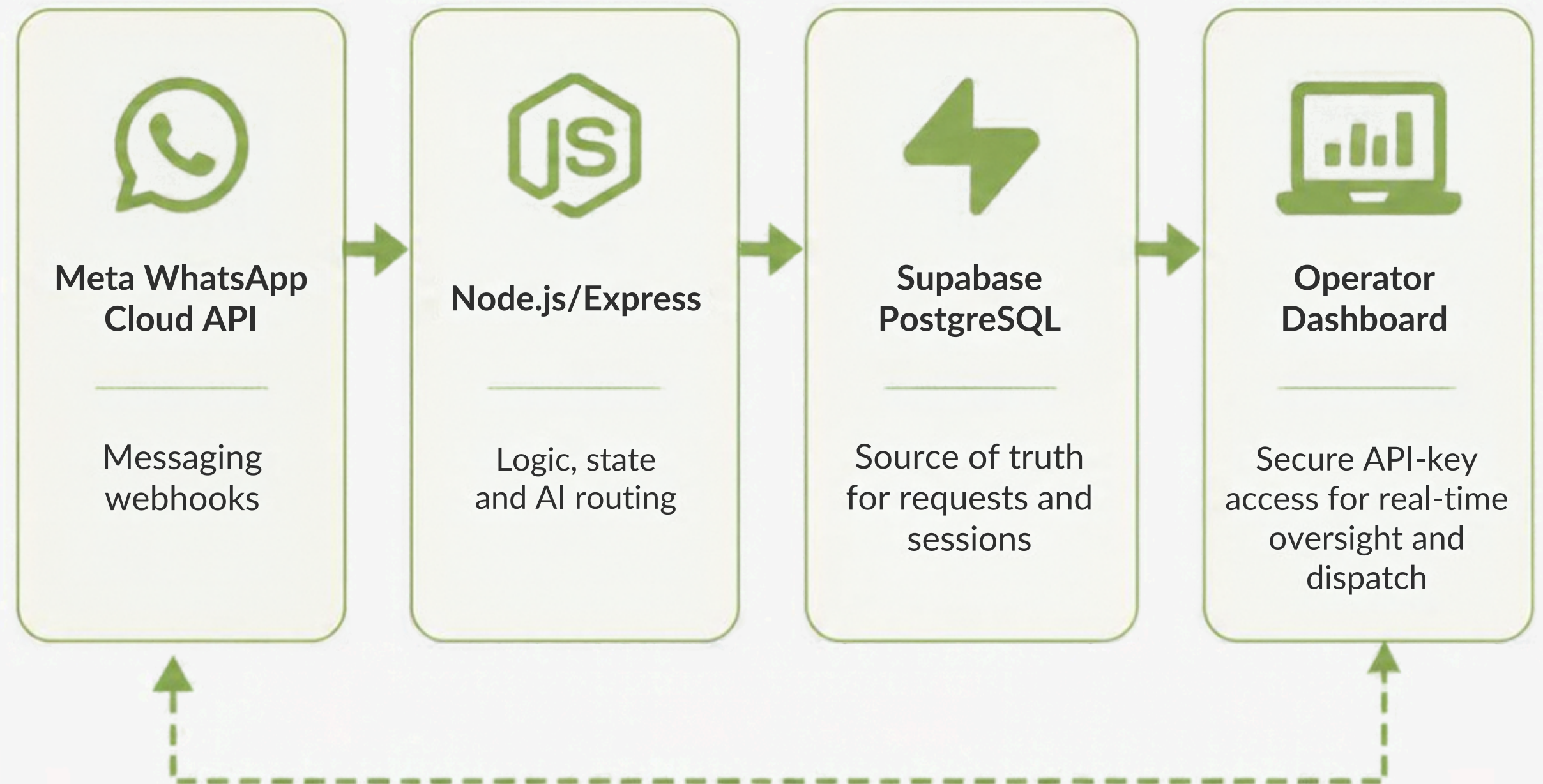
A resilient, event-driven backend connecting
Meta WhatsApp, Node.js, Supabase, and Qwen AI.

Meta WhatsApp Cloud API for
messaging webhooks

Node.js/Express managing logic,
state, AI routing

Supabase PostgreSQL as source of
truth for requests and sessions

Secure API-key-protected dashboard
for real-time oversight and dispatch



Strategic Trade-offs

Prioritized rapid end-to-end **validation** over premature **optimization**, with a clear path to production scaling.



Dashboard Filtering

Client-side for instant UI feedback; server pagination planned for scale.



Webhook Deduplication

In-memory Node.js Map for speed; Redis planned for horizontal scaling.



Communication

WhatsApp-only MVP; SMS fallback (Africa's Talking) mapped for rural coverage.



Infrastructure

Render free tier for zero-cost validation; migration to auto-scaling containers planned.

Future Roadmap

Transitioning from validated MVP to a **highly available, multi-channel** production system.

